

AI-Based Smart Window Control System for Sustainable Home

Project Description:-

This project proposes an AI-driven smart window management system that automatically recommends or controls window opening and closing based on environmental conditions such as indoor/outdoor temperature, humidity, air quality (AQI), rainfall probability, wind speed, and user preferences. The objective is to reduce energy consumption, improve indoor comfort, and promote sustainable living.

• Title:-

AI-Based Smart Window Control System for Sustainable Homes

• Student Name & College:-

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College name:- Chandigarh University

• SDG Alignment:-

Primary:

SDG 7 – Affordable and Clean Energy.

Secondary:

SDG 11 – Sustainable Cities and Communities SDG 13 – Climate Action.

• Problem Statement:-

How might we use Artificial Intelligence to intelligently monitor environmental conditions and automatically control household windows so that homes become more energy-efficient, environmentally sustainable, and comfortable for **occupants**.

• AI Solution Overview:-

The **AI-Based Smart Window Control System for Sustainable Homes** is an intelligent software solution developed to improve household energy efficiency, indoor comfort, and environmental sustainability by automating window management through Artificial Intelligence (AI). The system continuously analyzes environmental conditions such as indoor temperature, outdoor temperature, humidity, air quality index (AQI), rainfall probability, wind speed, weather

conditions, time of day, and user preferences to determine the optimal window position. By making intelligent decisions, the system reduces unnecessary energy consumption caused by inefficient manual window operation and minimizes the dependence on heating, ventilation, and air conditioning (HVAC) systems.

The AI solution follows a structured workflow consisting of **data collection, validation, preprocessing, pattern detection, classification, prediction, decision support, explainable AI, and sustainability analysis**. Initially, environmental data is collected through manual user input or weather APIs. The data is validated to ensure accuracy and then preprocessed to generate meaningful features such as temperature differences and time-based environmental conditions. The AI engine detects patterns, classifies the environmental state, and predicts whether the window should be **opened, closed, partially opened, or remain unchanged**. Each recommendation is accompanied by a confidence score and a clear explanation, allowing users to understand the reasoning behind the AI's decision.

The system incorporates **Prompt Engineering** and supports **IBM Granite Models** or other Large Language Models (LLMs) to generate natural language explanations, personalized energy-saving recommendations, and sustainability insights. An **Agentic AI workflow** continuously monitors environmental conditions, updates recommendations in real time, and automatically refreshes the dashboard. The solution also estimates energy savings, carbon emission reductions, monthly electricity cost savings, and an overall sustainability score.

To ensure responsible AI usage, the system follows the principles of **transparency, fairness, privacy, and human oversight**. Users can always review or override AI recommendations. By intelligently managing household windows, the proposed solution contributes to **SDG 7 – Affordable and Clean Energy**, while also supporting **SDG 11 – Sustainable Cities and Communities** and **SDG 13 – Climate Action** through improved energy efficiency, reduced carbon emissions, and sustainable home management.

• **Target Users:-**

1. **Homeowners:** Individuals who want to reduce electricity consumption, improve indoor comfort, and automate window management based on changing environmental conditions.

2. **Apartment Residents:** People living in apartments or residential complexes who wish to maintain a comfortable indoor environment while minimizing energy usage and reducing utility costs.
3. **Smart Home Users:** Individuals who use or plan to adopt smart home technologies and want an AI-powered solution that integrates intelligent window control with existing home automation systems.
4. **Property Developers and Builders:** Residential builders and smart city developers looking to incorporate sustainable and energy-efficient technologies into modern buildings and housing projects.
5. **Educational Institutions:** Schools, colleges, and universities interested in demonstrating AI, sustainability, and smart home technologies for research, learning, and practical applications.
6. **Researchers and AI Developers:** Researchers working in Artificial Intelligence, Internet of Things (IoT), sustainable technologies, and energy management who can use the system for further development and innovation.
7. **Environmental Organizations:** Organizations promoting energy conservation, climate action, and sustainable living can use the system to demonstrate how AI contributes to reducing carbon emissions and improving environmental efficiency.
8. **Government and Smart City Authorities:** Municipalities and government agencies implementing smart city initiatives can adopt the system to encourage energy-efficient residential infrastructure and support sustainable urban development.

- **Responsible AI Considerations:-**

The **AI-Based Smart Window Control System for Sustainable Homes** is designed by following the principles of **Responsible Artificial Intelligence (AI)** to ensure transparency, fairness, privacy, reliability, and user control. The system analyzes environmental parameters such as temperature, humidity, air quality, rainfall probability, and wind speed to recommend the optimal window position for improving energy efficiency and indoor comfort.

To maintain **transparency**, every AI recommendation is accompanied by a clear explanation of the environmental factors influencing the decision. This helps users understand why the system suggests opening, closing, or partially opening the window.

The system follows the principle of **human oversight**, allowing users to accept, reject, or manually override AI recommendations. The final decision always remains with the user.

To protect **privacy**, only environmental data required for analysis is processed. Personal or sensitive information is neither collected nor stored. If weather APIs are used, location access is requested only with user permission.

The AI ensures **fairness and reliability** by making recommendations solely based on environmental conditions, providing consistent and unbiased results for all users. The system is tested using various weather scenarios to improve prediction accuracy.

By promoting intelligent energy management and reducing unnecessary electricity consumption, the project supports **SDG 7 – Affordable and Clean Energy**, while also contributing to **SDG 11 – Sustainable Cities and Communities** and **SDG 13 – Climate Action**.

• **Expected Impact:-**

1. Reduce unnecessary electricity consumption through intelligent window management.
2. Improve energy efficiency by minimizing the use of air conditioning and heating systems.
3. Enhance indoor comfort by maintaining optimal ventilation based on environmental conditions.
4. Improve indoor air quality by considering Air Quality Index (AQI) and weather conditions.
5. Reduce carbon emissions by promoting energy-efficient home management.
6. Provide real-time AI recommendations for opening, closing, or partially opening windows.
7. Increase user awareness of sustainable living and smart home technologies.
8. Support homeowners in making informed decisions through Explainable AI recommendations.

9. Enable software-based automation without requiring complex hardware during the prototype stage.
10. Provide a scalable solution that can be integrated with IoT devices and smart home systems in the future.
11. Demonstrate the practical application of Artificial Intelligence for solving real-world environmental challenges.
12. Support **SDG 7 – Affordable and Clean Energy** by improving household energy efficiency.
13. Contribute to **SDG 11 – Sustainable Cities and Communities** by encouraging intelligent home automation.
14. Promote **SDG 13 – Climate Action** by reducing energy wastage and lowering greenhouse gas emissions.
15. Create a user-friendly, reliable, and sustainable AI solution that can be adopted in modern homes and future smart city initiatives.

Prototype:-

• Prompt Workflows:-

The workflow ensures that environmental data is processed systematically before an AI model generates an explainable response.

Step 1: Environmental Data Collection

The system collects real-time environmental data either from manual user input or weather APIs. The collected parameters include:

- Indoor Temperature
- Outdoor Temperature
- Humidity
- Air Quality Index (AQI)
- Rainfall Probability
- Wind Speed
- Weather Condition
- Time of Day
- User Preference (Optional)

Step 2: Data Validation

The collected values are validated to ensure they are complete and within acceptable ranges. Invalid or missing values are identified before AI processing begins.

Step 3: Prompt Generation

The validated environmental data is converted into a structured prompt for the AI model.

Example Prompt:

"Analyze the following environmental conditions: Indoor Temperature = 24°C, Outdoor Temperature = 34°C, Humidity = 68%, AQI = 95, Rain Probability = 20%, Wind Speed = 12 km/h, Weather = Cloudy. Determine whether the household window should be Open, Close, Partially Open, or Remain Unchanged. Explain the reasons for your recommendation, estimate the energy-saving benefits, and provide sustainability advice."

Step 4: AI Analysis

The AI model analyzes the environmental parameters, detects patterns, classifies the current conditions, predicts the optimal window position, and generates an explainable recommendation.

Step 5: AI Response

The AI returns:

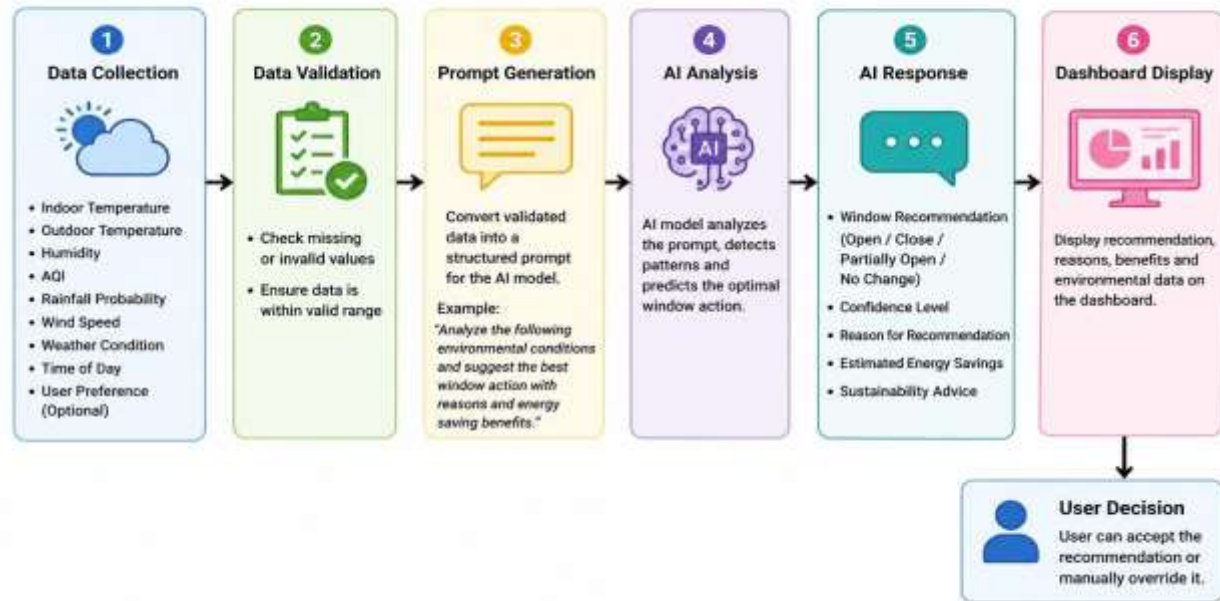
- Window Recommendation (Open / Close / Partially Open)
- Confidence Level
- Reason for Recommendation
- Estimated Energy Savings
- Sustainability Suggestion

Step 6: Dashboard Display

The recommendation and explanation are displayed on the Streamlit dashboard, allowing users to review the AI decision and manually accept or override the recommendation if required.

Workflow Summary:

Prompt Workflow



• RAG demo:-

When environmental data such as indoor temperature, outdoor temperature, humidity, air quality (AQI), rainfall probability, and wind speed is received, the system searches the knowledge base for the most relevant information. For example, if the AQI is high or heavy rainfall is expected, the system retrieves predefined recommendations related to indoor air quality and weather safety. This retrieved information is then combined with the current environmental data and included in the prompt sent to the AI model.

The AI analyzes both the real-time environmental conditions and the retrieved knowledge to generate an intelligent recommendation. The response includes the suggested window action (Open, Close, or Partially Open), an explanation of the decision, expected energy-saving benefits, and sustainability advice.

RAG Workflow

Environmental Data → Retrieve Relevant Knowledge → Combine Context → AI Analysis → Window Recommendation → Dashboard Display

Example

Input

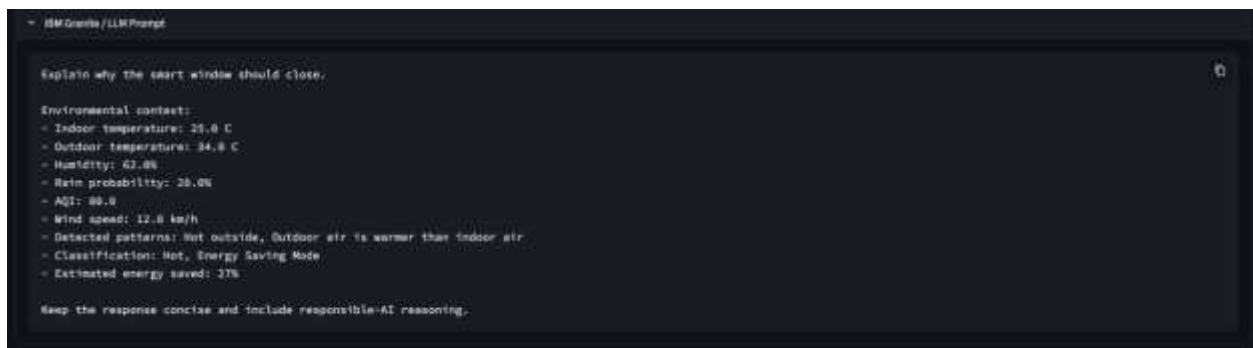
- Indoor Temperature: 24°C
- Outdoor Temperature: 35°C
- AQI: 185
- Rain Probability: 80%

Retrieved Knowledge

- High AQI may reduce indoor air quality if windows remain open.
- High rainfall probability indicates windows should remain closed.
- Closing windows during hot weather improves air-conditioning efficiency.

AI Output

- **Recommendation:** Close the Window
- **Reason:** Poor outdoor air quality and high rainfall probability may reduce indoor comfort and increase energy loss.
- **Expected Benefit:** Improved indoor air quality, reduced cooling energy consumption, and enhanced household energy efficiency.



• Agent logic:-

The **AI-Based Smart Window Control System** uses an intelligent AI agent to continuously monitor environmental conditions and provide real-time recommendations for window management.

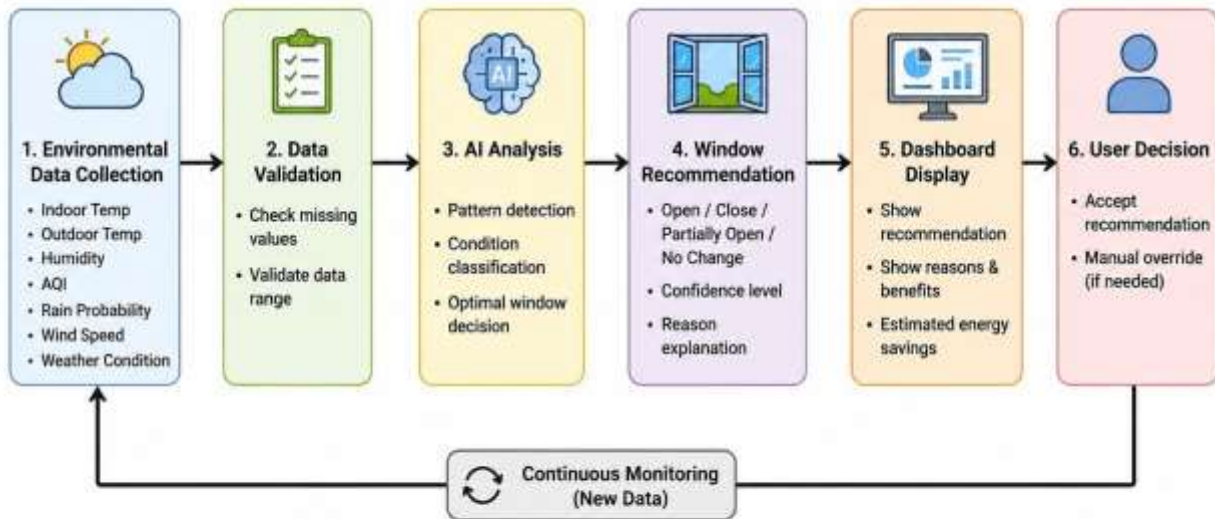
Agent Workflow

1. **Collect Environmental Data**
 - a. Read indoor temperature.

- b. Read outdoor temperature.
 - c. Read humidity.
 - d. Read Air Quality Index (AQI).
 - e. Read rainfall probability.
 - f. Read wind speed.
 - g. Read weather conditions and time of day.
- 2. Validate Data**
- a. Check whether all input values are complete and within valid ranges.
 - b. Discard or request correction for invalid data.
- 3. Analyze Environmental Conditions**
- a. Detect patterns such as hot weather, poor air quality, heavy rain, or strong winds.
 - b. Classify the current environmental state.
- 4. Generate AI Recommendation**
- a. Predict the optimal window action:
 - i. Open Window**
 - ii. Close Window**
 - iii. Partially Open Window**
 - iv. No Change**
 - b. Calculate the confidence level of the recommendation.
- 5. Provide Explainable Output**
- a. Display the reason for the recommendation.
 - b. Estimate energy savings and environmental benefits.
 - c. Suggest sustainable actions to the user.
- 6. Update Dashboard**
- a. Display the AI recommendation, environmental readings, and system status on the Streamlit dashboard.
 - b. Allow the user to accept or manually override the recommendation.
- 7. Continuous Monitoring**
- a. Repeat the monitoring and decision-making process at regular intervals or whenever new environmental data becomes available.

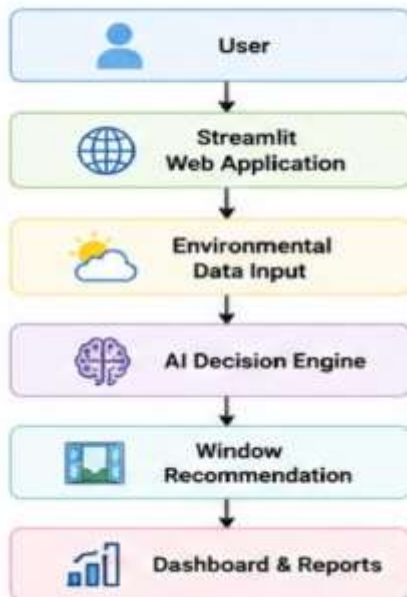
Agent Logic Flow

AI-Based Smart Window Control System Agent Logic Flow

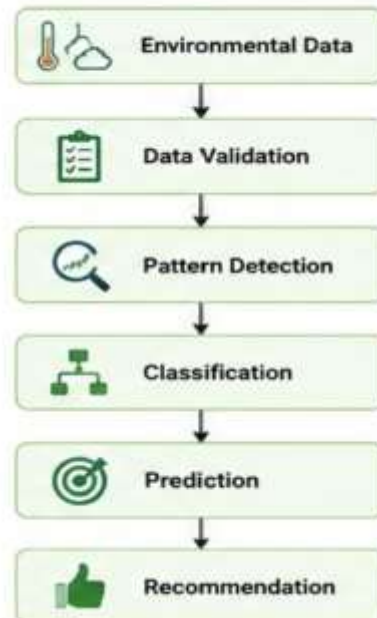


• Flow Diagrams:-

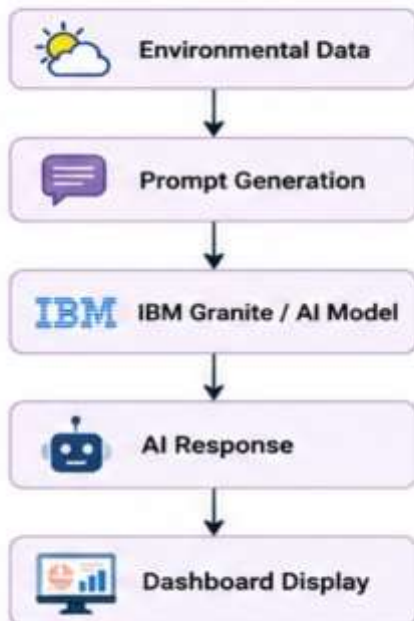
1 Overall System Workflow



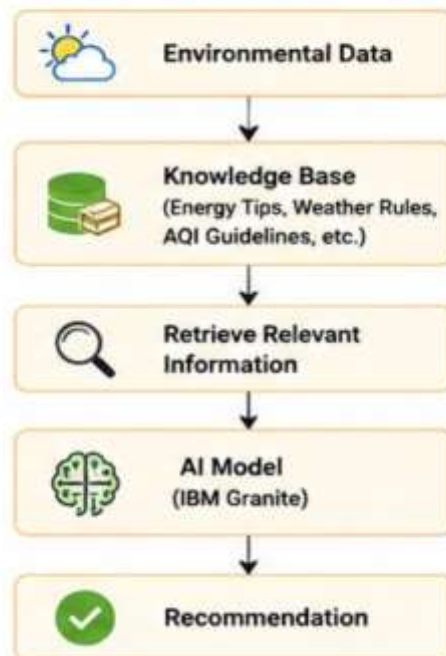
2. AI Workflow



3 Prompt Workflow

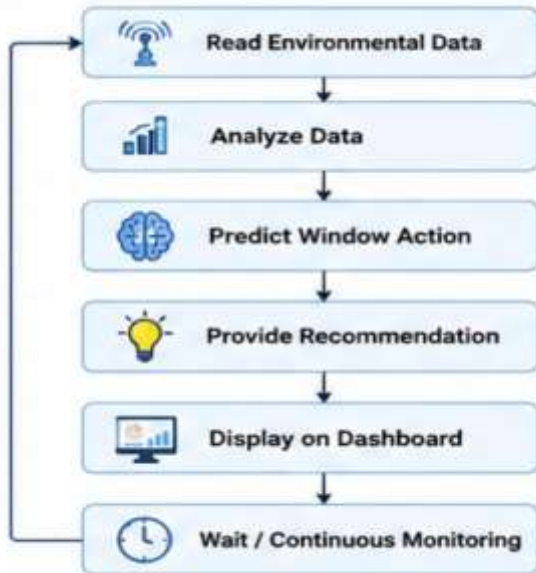


4 RAG Workflow (Retrieval-Augmented Generation)



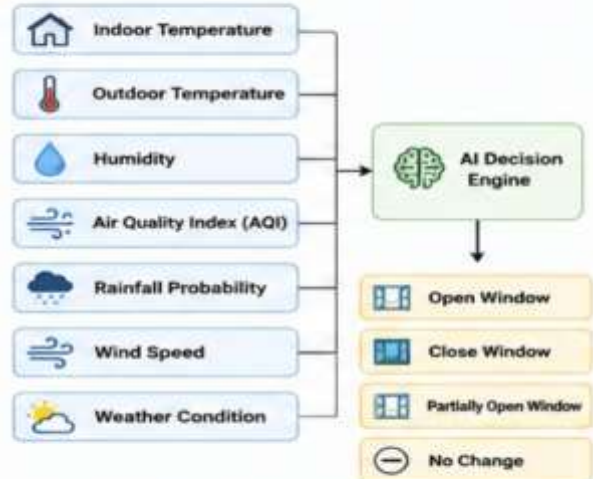
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Agent Workflow



6

Window Decision Flow



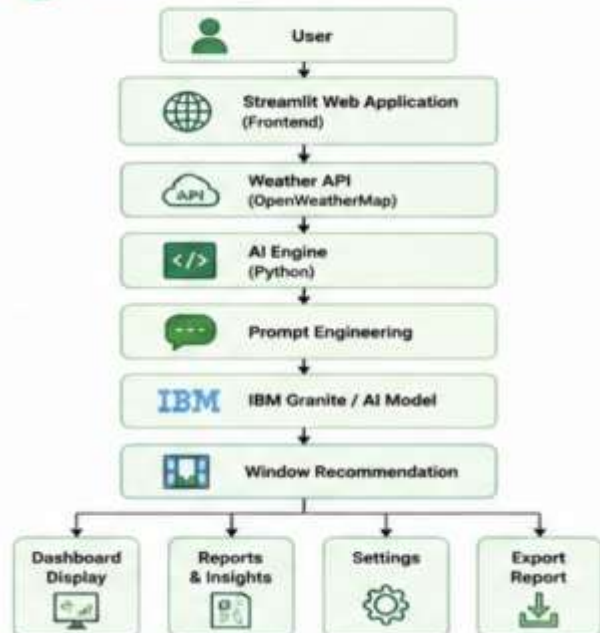
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Dashboard Flow

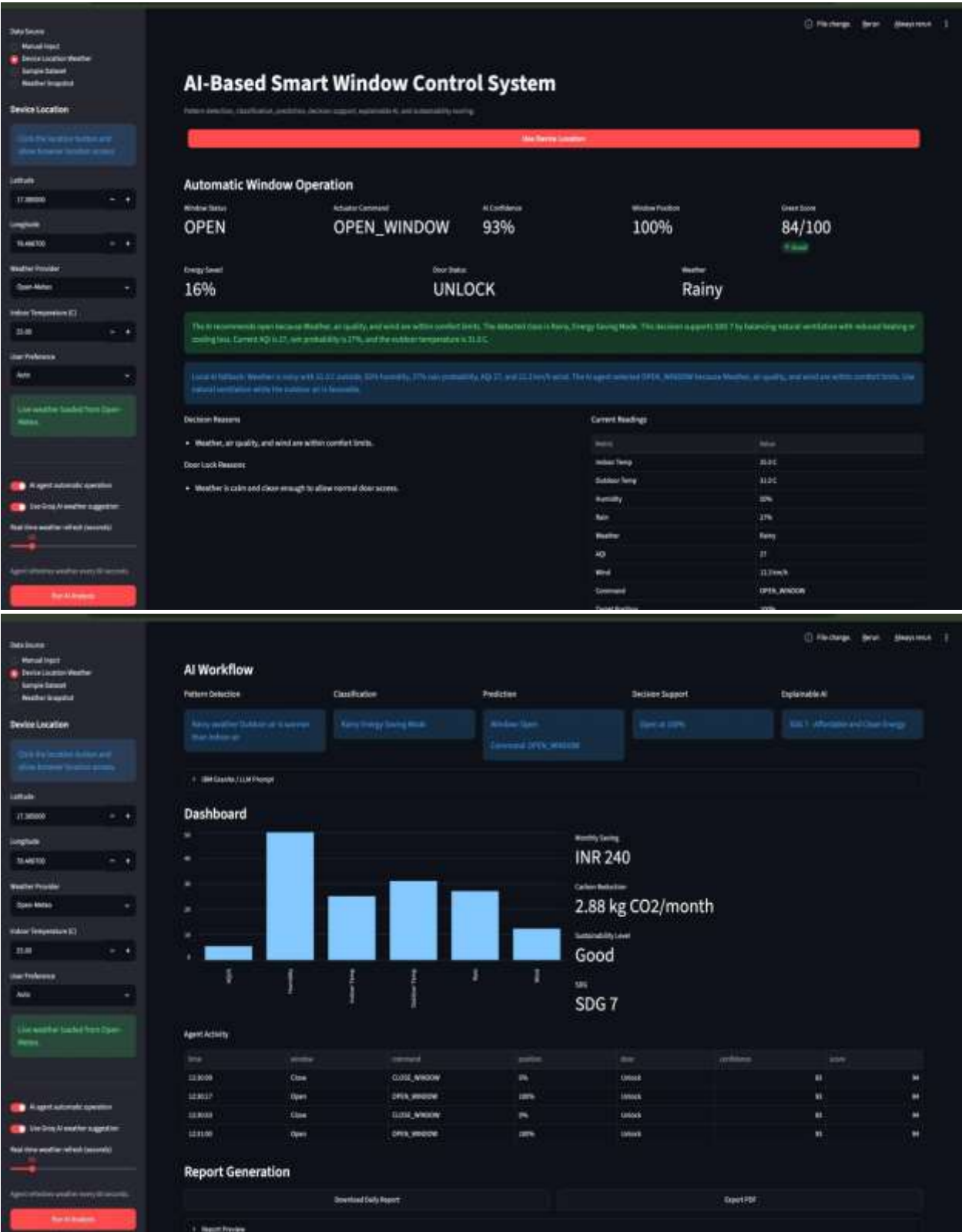


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Complete System Architecture



• Screen Shots:-



```
smart_window_daily_report (1).txt
AI-Based Smart Window Control System Report
Generated: 2026-06-25 12:12

Environmental Data
Indoor Temperature: 25.0 C
Outdoor Temperature: 34.0 C
Humidity: 62%
Rain Probability: 20%
AQI: 80
Wind Speed: 12.0 km/h
Time of Day: 12:12
User Preference: Auto

AI Workflow Output
Patterns: Hot outside, Outdoor air is warmer than indoor air
Classification: Hot, Energy Saving Mode
Prediction: Close
Recommendation: Close
Door Action: Unlock
Door Reasons: Weather is calm and clean enough to allow normal door access.
Confidence: 83%

Explainable AI
The AI recommends close because Outdoor temperature is high, so closing reduces cooling loss.
The detected class is Hot, Energy Saving Mode. This decision supports SDG 7 by balancing natural
ventilation with reduced heating or cooling loss. Current AQI is 80, rain probability is 20%,
and the outdoor temperature is 34.0 C.

Sustainability
Estimated Energy Saved: 27%
```

Impact Statement:-

•What changes if this solution is implemented?

1. Reduces unnecessary electricity consumption by optimizing window operations.
2. Minimizes dependence on air conditioners and heating systems.
3. Improves indoor air quality through intelligent ventilation management.
4. Maintains a comfortable indoor environment based on real-time weather conditions.
5. Provides AI-powered recommendations for opening, closing, or partially opening windows.
6. Reduces carbon emissions by promoting energy-efficient practices.
7. Increases awareness of sustainable and eco-friendly living.
8. Supports smart home automation using Artificial Intelligence.
9. Enables data-driven decision-making through real-time environmental analysis.
10. Can be integrated with IoT devices for fully automated smart home systems in the future.

11. Encourages responsible use of AI in everyday household applications.
12. Contributes to **SDG 7 (Affordable and Clean Energy)**, **SDG 11 (Sustainable Cities and Communities)**, and **SDG 13 (Climate Action)**.

- **Who benefits and how?**

1. Homeowners
 - a. Lower electricity bills.
 - b. Improved indoor comfort.
 - c. Better ventilation and healthier living conditions.
2. Apartment Residents
 - a. Enhanced indoor air quality.
 - b. Reduced energy consumption.
 - c. Smart recommendations for window management.
3. Smart Home Users
 - a. AI-based automation for window control.
 - b. Better integration with smart home technologies.
 - c. Improved energy efficiency.
4. Educational Institutions
 - a. Practical demonstration of AI and sustainability concepts.
 - b. Useful for research, internships, and student projects.
5. Researchers & AI Developers
 - a. Platform for developing advanced AI and IoT applications.
 - b. Opportunity to improve intelligent decision-making systems.